

Problem 2

From a table of random numbers, two numbers are randomly written. Let A denote the event that one of these numbers is prime and B that one of them is even. What do the events AB and $A + B$ represent?

Solution

We will interpret the events AB and $A + B$ in terms of the properties of the two randomly selected numbers.

Setup

Let the two randomly selected numbers be denoted as n_1 and n_2 .

Event A : At least one of the two numbers is prime.

$$A = \{\text{at least one of } n_1, n_2 \text{ is prime}\}$$

Event B : At least one of the two numbers is even.

$$B = \{\text{at least one of } n_1, n_2 \text{ is even}\}$$

Part 1: Event AB (Intersection)

Definition

$AB = A \cap B$ represents the event that both A and B occur simultaneously.

Interpretation

$$AB = \{\text{at least one number is prime AND at least one number is even}\}$$

What Outcomes Are in AB ?

The event AB occurs when:

- At least one of $\{n_1, n_2\}$ is prime, AND
- At least one of $\{n_1, n_2\}$ is even

Possible Cases

This can happen in several ways:

1. Both numbers satisfy both properties: One number is both prime and even (which means it's 2), and the other is anything that makes the conditions true.
2. One number is prime (and odd), and the other number is even (and not necessarily prime).
3. One number is prime and even (i.e., 2), and the other is anything.
4. Both numbers have the required properties in different combinations.

Clearer Formulation

AB occurs in the following cases:

- One number is 2 (which is both prime and even), and the other is any number.
- One number is an odd prime, and the other is an even number (possibly 2, or any other even number).
- One number is an odd prime, and the other is 2.
- Both numbers are 2.

Simplified Statement

$$AB = \{\text{At least one number is prime and at least one number is even}\}$$

More explicitly: The two numbers include at least one prime and at least one even number (these could be the same number if it's 2, or different numbers).

Part 2: Event $A + B$ (Union)

Definition

$A + B = A \cup B$ represents the event that at least one of A or B occurs.

Interpretation

$$A + B = \{\text{at least one number is prime OR at least one number is even (or both)}\}$$

What Outcomes Are in $A + B$?

The event $A + B$ occurs when at least one of the following is true:

- At least one of $\{n_1, n_2\}$ is prime
- At least one of $\{n_1, n_2\}$ is even

Complement Approach

It's easier to describe what's NOT in $A + B$:

$$\overline{A + B} = \overline{A} \cap \overline{B}$$

where:

- $\overline{A}$ = "neither number is prime" = "both numbers are composite (or 1)"
- $\overline{B}$ = "neither number is even" = "both numbers are odd"

Therefore:

$$\overline{A + B} = \{\text{both numbers are odd composite numbers (or 1)}\}$$

Simplified Statement

$$A + B = \{\text{At least one number is prime or at least one number is even}\}$$

Equivalently: It is NOT the case that both numbers are odd composite numbers.

Examples

Example for AB

- (2, 7): AB occurs (2 is prime and even, 7 is prime and odd)
- (3, 8): AB occurs (3 is prime, 8 is even)
- (2, 4): AB occurs (2 is prime and even, 4 is even)
- (3, 5): AB does NOT occur (both prime but neither even)
- (4, 6): AB does NOT occur (both even but neither prime)
- (9, 15): AB does NOT occur (neither prime nor even)

Example for $A + B$

- (2, 7): $A + B$ occurs (both are prime, and one is even)
- (3, 5): $A + B$ occurs (both are prime)
- (4, 6): $A + B$ occurs (both are even)
- (9, 15): $A + B$ does NOT occur (both are odd composite)
- (1, 9): $A + B$ does NOT occur (both are odd and neither prime)

Final Answer**Event AB :**

$$AB = \{\text{At least one of the two numbers is prime AND at least one is even}\}$$

This occurs when the pair includes:

- At least one prime number (possibly 2 or an odd prime), AND
- At least one even number (possibly 2 or another even number)

Event $A + B$:

$$A + B = \{\text{At least one number is prime OR at least one number is even}\}$$

Equivalently: NOT both numbers are odd composite numbers.

In set notation:

- $AB = A \cap B$: Both conditions must be satisfied
- $A + B = A \cup B$: At least one condition must be satisfied